



RUPESH RAO

Senior DevOps Engineer

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Professional Summary:

- Results-Driven **Senior DevOps Engineer** with **over 10 years** of experience in Cloud Engineering, DevOps, DevSecOps and Site Reliability Engineering. Adept at architecting, automating and optimizing mission-critical cloud infrastructures across AWS, Azure and hybrid environments. Skilled in driving cloud transformation initiatives with a focus on CI/CD, Infrastructure as Code (IaC), cloud security, orchestration and cost optimization.
- Skilled in designing, deploying and scaling infrastructure using **AWS services** such as **EKS, EC2, S3, Lambda, IAM, Secrets Manager, SNS, SQS, RDS** and **API Gateway**.
- Experienced in managing **Azure** environments with services including **AKS, Azure DevOps, Azure Functions, Blob Storage, Key Vault, App Services** and **Azure API Management**.
- Proficient in Infrastructure as Code (IaC) using **Terraform Enterprise, Pulumi, Ansible, AWS CloudFormation** and **Azure Resource Manager (ARM)** templates for provisioning cloud resources.
- Expertise in building robust CI/CD pipelines using **Jenkins, Azure DevOps, GitHub Actions, AWS CodePipeline** and **Bitbucket**, improving deployment reliability and frequency.
- Skilled in implementing **GitOps workflows** using **ArgoCD, FluxCD** and **Kustomize** for declarative, automated Kubernetes application deployments.
- Hands-on experience with **Kubernetes orchestration** using **AKS, EKS, GKE, Docker, containerd** and **CRI-O**, ensuring high availability and microservices scalability.
- Proficient in **Kubernetes** package management with **Helm** and **Kustomize** and implementing service meshes such as **Istio** and **Linkerd** for secure and resilient communication between services.
- Strong knowledge of **DevSecOps tools**, integrating **SonarQube, Trivy, OWASP ZAP, Prisma Cloud, Twistlock, Cyscale, Veracode** and **OWASP Dependency Check** into CI/CD workflows.
- Adept at enforcing security best practices through **Zero Trust security models, OAuth, RBAC, IAM Policies** and **Policy-as-Code** using **OPA (Open Policy Agent)** and **Sentinel**.
- Experienced in generating and managing **Software Bill of Materials (SBOM)** using **Syft** and **Grype**, ensuring transparency and security in the software supply chain.
- Skilled in secrets and credential management using **HashiCorp Vault, AWS Secrets Manager** and **Azure Key Vault**, enforcing encryption and access control.
- Extensive monitoring and observability experience using tools like **Prometheus, Grafana, Datadog, ELK Stack, Splunk, Dynatrace, New Relic** and **CloudWatch**.
- Configured **Kube-Prometheus Stack, Loki** and **Fluentd** for Kubernetes-native observability and cluster health monitoring.

- Strong scripting and automation skills in **Python, Bash, Groovy, Shell, YAML** and **HCL**, used to automate deployments, configuration management and cloud operations.
- Proficient in managing scalable and secure APIs using **AWS API Gateway, Azure API Management, OAuth-based authentication** and **load balancers** including **ALB, NLB** and **Azure Front Door**.
- Demonstrated success in optimizing cloud spends using **AWS Cost Explorer, Azure Cost Management** and **automated FinOps strategies**, achieving up to 20% cost reduction.
- Experienced with version control and collaboration using **GitHub, Bitbucket, Azure Repos** and managing Git branching strategies, access controls and Webhook-triggered workflows.
- Proven ability to implement end-to-end automated CI/CD and **GitOps pipelines**, reducing deployment failures by 40% and increasing delivery velocity by 60%.

Professional Experience:

Client:	CBRE,	Dallas,	TX
			Sep
t 2024 - Till Now			
Senior DevOps Engineer			

Project Description: As a Senior DevOps Engineer at CBRE, I led the modernization, automation and security transformation of the Sequentra lease abstraction management platform. My role focused on orchestrating the migration of workloads from AWS to Azure, implementing robust cloud-native infrastructure and CI/CD pipelines and integrating industry-leading security and observability tools. This initiative aimed to boost scalability, reliability and compliance for enterprise-grade lease management applications.

Responsibilities:

- Spearheaded the end-to-end migration of infrastructure and applications from AWS to Azure using **AWS Migration Hub, AWS DMS** and **Azure Migrate**, ensuring zero-downtime transitions and minimal disruption.
- Architected and deployed secure, auto-scaling **Azure Kubernetes Service (AKS)** clusters with optimized networking, service mesh policies (Istio) and integrated RBAC for governance.
- Developed Infrastructure-as-Code (IaC) frameworks using **Terraform Enterprise, Pulumi** and **Ansible** to provision scalable Azure resources, including **VMs, App Services, AKS, Key Vault, Blob Storage** and databases.
- Engineered GitOps-powered CI/CD pipelines leveraging **Azure DevOps, GitHub Actions, Jenkins, ArgoCD** and **FluxCD**, accelerating delivery cycles by 50% and minimizing manual interventions.
- Enabled Kubernetes-native deployments using **Docker, Helm** and **ArgoCD**, orchestrating automated deployments of microservices and ensuring consistent delivery across dev, staging and production.
- Integrated serverless components using **AWS Lambda** and **API Gateway**, optimizing compute efficiency for event-driven Node.js applications.
- Achieved 99.99% application availability with proactive monitoring through **Datadog, Prometheus, Grafana** and **AI-based anomaly detection**, coupled with automated infrastructure healing.

- Strengthened DevSecOps maturity by embedding security into the CI/CD process via **Snyk, Veracode, Prisma Cloud, Cyscale** and **OWASP ZAP**.
- Instituted Software Bill of Materials (SBOM) practices using **Syft** and **Grype** to mitigate supply chain vulnerabilities and maintain artifact transparency.
- Designed and enforced Zero Trust security models with **Azure AD, OAuth, RBAC, IAM Policies** and **API authentication**, increasing compliance posture by 40%.
- Deployed **Policy-as-Code** controls using **OPA (Open Policy Agent)** and **Sentinel (Terraform Enterprise)** to implement automated governance in multi-cloud environments.
- Established comprehensive secrets management solutions via **HashiCorp Vault, AWS Secrets Manager** and **Azure Key Vault**, ensuring secure, encrypted access to sensitive credentials.
- Optimized infrastructure costs and cloud resource utilization through FinOps practices using **AWS Cost Explorer, Azure Cost Management** and automated rightsizing—resulting in 20% operational cost reduction.
- Secured and accelerated Kubernetes communications by deploying **Istio**-based service mesh, **RBAC policies** and encryption across microservices.
- Deployed security monitoring tools (**SonarQube, Trivy, OWASP ZAP**) to enable real-time detection of vulnerabilities and ensure adherence to secure coding standards.
- Architected resilient networking solutions across cloud environments using **VPCs, DNS, AWS ALB/NLB, Azure Front Door, VPNs** and **firewall configurations**.
- Implemented **Imperva WAF**, data masking and database security controls to protect sensitive lease-related data and ensure regulatory compliance.
- Built comprehensive observability dashboards and alerts using **Prometheus, Grafana, Datadog, ELK Stack, Splunk, Dynatrace** and **New Relic**, enabling real-time application health and performance tracking.
- Deployed **Kube-Prometheus Stack, Loki** and **Fluentd** for Kubernetes-native monitoring and log aggregation.
- Delivered scalable microservices using **Node.js** and **TypeScript**, leveraging **ECS Fargate, API Gateway, DynamoDB** and **AWS Lambda** for cloud-native workloads.
- Implemented scalable and secure API strategies via **AWS API Gateway, Azure API Management** and OAuth-based authentication.
- Automated backup, rollback and failover strategies for disaster recovery and high availability of mission-critical applications.
- Led ITSM workflow integration with **JIRA, ServiceNow** and **Confluence**, including on-call support and incident response for production environments and managed secure token rotation and credential management processes, mitigating risks for sensitive applications and services.

Client: Huntington Bank, Columbus, OH

Sep

2022 - Sep 2024

Senior DevOps Engineer

Project Description: At Huntington Bank, I served as a Senior DevOps Engineer responsible for modernizing legacy infrastructure and migrating mission-critical banking applications from on-premise data centers to the Microsoft Azure cloud. I implemented scalable, secure, and cost-effective Azure-native infrastructure and CI/CD pipelines, significantly improving deployment

speed, cloud reliability, compliance, and real-time analytics across the enterprise.

Responsibilities:

- Led the migration of legacy banking applications from on-premise infrastructure to **Azure Cloud**, utilizing **Azure Migrate**, **Azure Database Migration Service**, and **Cloud Migration Factory (CMF)** to ensure zero-downtime and data integrity.
- Architected scalable **Azure infrastructure** leveraging **Azure AD**, **Virtual Machines**, **Azure Blob Storage**, **Azure App Services**, **Azure SQL**, **Azure Functions**, and **Azure Event Grid**, achieving improved performance and cost reduction.
- Designed and provisioned Kubernetes clusters using **Azure Kubernetes Service (AKS)** and **OpenShift**, implementing Infrastructure-as-Code using **Terraform**, **Pulumi**, and **Ansible** to ensure high availability and automation.
- Deployed microservices using **Helm charts** and **Kubernetes manifests**, implementing **RBAC**, persistent volumes, namespaces, and load balancers to ensure secure and compliant cluster operations.
- Integrated **OpenShift** as a container orchestration platform, enforcing enterprise RBAC policies and supporting hybrid deployments.
- Automated CI/CD pipelines using **Jenkins**, **Bitbucket**, **Azure DevOps**, and **Ansible**, reducing manual effort and improving deployment consistency across environments.
- Embedded DevSecOps best practices by integrating security tools such as **SonarQube**, **Twistlock**, **Prisma Cloud**, **OWASP ZAP**, and **OWASP Dependency Check** into the pipeline for continuous vulnerability scanning.
- Managed full container lifecycle with **Docker**, **JFrog Artifactory**, **Docker Hub**, **Kubernetes**, and **OpenShift**, streamlining image repository management and deployment workflows.
- Secured credentials and secrets using **HashiCorp Vault** and **Azure Key Vault**, implementing role-based and encrypted access to sensitive application data.
- Applied **Site Reliability Engineering (SRE)** principles, achieving 99.9% uptime by deploying **SLIs/SLOs**, **auto-healing mechanisms**, and real-time health checks.
- Automated infrastructure provisioning using **Terraform**, **Ansible**, and **Azure Resource Manager (ARM) templates**, enhancing deployment speed and standardization.
- Implemented **FinOps** strategies using **Azure Cost Management**, rightsizing resources, and enforcing cost governance policies to reduce cloud spend and improve financial visibility.
- Built scalable, real-time **ETL pipelines** using **Apache Spark**, **Databricks**, and **Azure Data Factory**, integrating with **Blob Storage**, **Cosmos DB**, **Kafka**, and **Snowflake** for advanced analytics and processing.
- Deployed and maintained **.NET enterprise applications** using **Azure App Services** and **AKS**, while integrating full CI/CD automation with **Azure DevOps**.
- Established robust observability across systems using **Datadog**, **Splunk**, **Grafana**, **Prometheus**, **ELK Stack**, **Dynatrace**, **New Relic**, and **Azure Monitor**, enabling end-to-end visibility.
- Automated deployment and compliance processes through **Ansible Playbooks** and **Ansible Tower**, improving consistency and governance across environments.
- Managed and tuned **databases** (PostgreSQL, MySQL, MongoDB, Oracle DB) and **application middleware** (WebLogic, WebSphere, JBoss, Tomcat), ensuring optimized performance and uptime.
- Enhanced security and compliance through Azure-native controls and remediation

workflows, leading to a measurable increase in **Azure Security Center** score.

- Created automation and orchestration scripts using **Python**, **Shell**, and **Groovy**, improving DevOps efficiency and infrastructure repeatability.
- Delivered multiple **Proof-of-Concept (PoC)** solutions using **Azure Compute**, **Blob Storage**, **Web & Mobile Services**, **Azure Data Factory**, **SQL Data Warehouse**, and **Resource Groups**, validating emerging technologies and business use cases.
- Integrated **Kafka monitoring** into **Datadog**, ensuring the availability and performance of real-time transaction data streams in production.

Client: WellPoint, Thousand Oaks, CA

Jun

2019- Aug 2022

Infrastructure Engineer

Project Description: As an Infrastructure Engineer at Wellpoint, I played a pivotal role in a large-scale cloud migration initiative, transitioning mission-critical services from Microsoft Azure to Google Cloud Platform (GCP). My responsibilities included end-to-end cloud infrastructure design, CI/CD automation, container orchestration and security hardening. I leveraged both Azure and GCP services to build scalable, secure and highly available environments, while enhancing automation, observability and cloud cost efficiency across enterprise healthcare applications.

Responsibilities:

- Led the migration of enterprise workloads from **Azure** to **GCP**, developing migration strategies and ensuring zero-downtime deployment of critical applications and services.
- Designed and deployed scalable **Azure infrastructure** components including **Virtual Machines**, **App Services**, **SQL Databases** and **Blob Storage**, optimizing for performance and cost-effectiveness.
- Managed and provisioned infrastructure-as-code (IaC) using **Terraform** across both **Azure** and **GCP**, enabling reproducible deployments of compute, storage, networking and Kubernetes clusters.
- Built and automated **CI/CD pipelines** using **Azure DevOps**, integrating with **Gradle**, **Kubernetes** and GCP tools to streamline build, test and deployment lifecycles.
- Deployed and managed **Azure Kubernetes Service (AKS)** and **Docker Swarm** clusters for orchestrating microservices, achieving scalability and resilience across environments.
- Implemented **Helm Charts** for packaging and deploying applications to Kubernetes, using **ChartMuseum** for Helm artifact management and version control.
- Integrated **Azure Active Directory (AAD)** for secure **IAM** across cloud resources and applications, ensuring compliance with enterprise security standards.
- Strengthened cloud security posture by configuring **Azure Security Center**, **Azure Sentinel** and **Aqua Security** for continuous threat monitoring and container security enforcement.
- Automated deployment of **Apache Spark** jobs via CI/CD, integrating with version control systems and IaC for repeatable big data processing workflows.
- Designed and deployed core **GCP infrastructure**, including **Compute Engine**, **Google Kubernetes Engine (GKE)**, **Cloud Storage** and **Cloud SQL**, to support migrated and cloud-native workloads.
- Developed automated build and deployment pipelines using **Google Cloud Build** and **Cloud**

Deployment Manager, enabling rapid feature delivery on GCP.

- Configured **GCP** networking components, including **VPC**, **Cloud Load Balancing** and **Cloud DNS**, ensuring high performance and availability of services.
- Executed seamless data migrations from **Azure SQL** and **Blob Storage** to **GCP Cloud SQL** and **Cloud Storage**, minimizing downtime and preserving data integrity.
- Designed and tested **disaster recovery (DR)** and **business continuity plans (BCP)** using **Google Cloud Storage** for automated backups and **Cloud Spanner** for global database availability.
- Implemented observability using **Google Cloud Monitoring** (formerly Stackdriver) integrated with third-party tools for real-time insights into infrastructure health and performance.
- Integrated **Dynatrace** with **Jenkins**, **Kubernetes** and CI/CD pipelines for comprehensive performance monitoring and incident detection.
- Deployed **New Relic** for application performance management and alerting, leading to a 25% reduction in deployment time and faster feature releases.

Client: First Bank, Denver, CO

July 2016 – May 2019

Cloud Engineer

Project Description: As a Cloud Engineer at First Bank, I played a key role in building and automating cloud-native infrastructure for banking applications in a highly regulated environment. My responsibilities spanned multi-tier architecture design, API management, infrastructure automation and DevOps integrations within AWS. I worked closely with cross-functional teams to ensure scalability, resilience and compliance through infrastructure-as-code, containerization and automated deployment strategies.

Responsibilities:

- Designed and deployed **multi-tier applications** on **AWS**, utilizing core services including **EC2**, **S3**, **RDS**, **DynamoDB**, **SNS**, **SQS**, **IAM** and **Route 53**, focusing on high availability and disaster recovery.
- Architected scalable and resilient **DNS solutions** using **AWS Route 53** and **DirectConnect**, improving traffic management and network performance for large-scale deployments.
- Automated infrastructure tasks with **Python scripts** integrated with the **AWS API**, reducing manual overhead and improving operational efficiency.
- Utilized **Chef** to automate EC2 instance configuration and server deployment, ensuring consistency and repeatability across test, staging and production environments.
- Implemented detailed monitoring and proactive alerting with **Amazon CloudWatch** for EC2 instances, significantly improving performance tuning and incident response.
- Collaborated with developers to design consistent deployment templates and workflows using **Spinnaker**, standardizing CI/CD processes across environments.
- Automated the provisioning and orchestration of cloud infrastructure using **AWS CloudFormation**, **Ruby scripts**, **Bamboo** and **AWS SAM (Serverless Application Model)**.
- Designed and deployed secure and scalable **RESTful APIs** using **AWS API Gateway**, **Lambda** and **CloudFormation** with **Swagger** specifications for documentation and integration.
- Managed API security, throttling and versioning through **API Gateway**, ensuring reliable and

compliant communication across distributed microservices.

- Managed **MongoDB** deployments using **Docker** containers and **MongoDB Atlas**, enabling high availability, scalability and managed backups in cloud environments.
- Implemented **containerization** strategies with **Docker** and orchestrated workloads using **Kubernetes** for development and testing environments, optimizing compute resource utilization and simplifying application rollouts.

Client: Edwards Life sciences Corp, Irvine, CA

Jan 2013 – June 2016

Build & Release Engineer

Responsibilities:

- Designed and implemented **CI pipelines** in **Jenkins**, integrating with **GitHub** and **Bitbucket**, enabling automated builds triggered by **webhooks** on code merges to the master branch.
- Managed **SVN** version control systems by administering branching strategies, tag management and user access policies to support team collaboration and controlled code promotion.
- Developed automated **build and deployment scripts** using **Maven** and **Ant** for Java applications, standardizing workflows across development, QA, staging and production.
- Integrated **Jenkins** with **Chef** to automate the deployment of **WAR** and **JAR** artifacts from **Nexus** to multiple **Apache Tomcat** environments.
- Configured and maintained **JBoss application servers**, including setup of **JDBC connections**, **JMS resources** and **security policies**, enhancing middleware performance and compliance.
- Utilized **JBoss Operations Network (JBoss ON)** for application inventory, health monitoring and proactive infrastructure insights.
- Integrated **JUnit** for unit testing and **SonarQube** for static code analysis within Jenkins pipelines to ensure automated test reporting and enforce code quality standards.
- Managed binary artifact repositories including **Nexus** and **JBoss Artifactory**, automating post-build uploads and version tracking of release components.
- Maintained build consistency by configuring Jenkins **build agents** and managing concurrent pipeline executions across shared nodes.
- Enforced build versioning and traceability standards to comply with **regulatory and audit requirements** in the healthcare domain.
- Automated deployment processes across environments with **parameterized Jenkins jobs**, enabling rapid hotfix rollouts and feature deployments.
- Participated in **release planning** and **change management**, coordinating with QA, developers and operations teams to minimize downtime during scheduled releases.
- Documented build and release processes in **Confluence**, including setup guides, troubleshooting workflows and post-release validation steps.

Client: Valuelabs, Hyderabad, India

Jun

2009 - Sep 2012

Software Engineer

Responsibilities:

- Developed comprehensive **UML diagrams** including **Use Case**, **Class**, **Sequence** and **Activity**

Diagrams to model and design application architecture based on business requirements.

- Built dynamic web interfaces using **JavaServer Pages (JSP)** and handled HTTP requests via **Servlets**, enabling seamless user interaction with backend systems.
- Deployed and maintained applications on **Apache Tomcat Server**, configuring deployment descriptors and ensuring application availability in production environments.
- Defined structured data flow using **XML** and **Document Type Definitions (DTD)** to facilitate data exchange and validation between components.
- Used **Sqoop** to import structured data from **Oracle** and other relational databases into data processing environments, enhancing integration and analytics capabilities.
- Developed backend logic and managed data access using **SQL Server**, including the creation of **Stored Procedures, Functions** and **Triggers** using **PL/SQL**.
- Designed and implemented intuitive GUIs with **JSP, HTML** and **DHTML**, while applying **JavaScript** for client-side validation and interactivity.
- Conducted **unit testing, bug tracking** and **issue resolution** throughout the development lifecycle, ensuring high-quality application releases.
- Applied **JavaBeans, JDBC** and **MS-Access** for data persistence and lightweight data access solutions in prototype and legacy modules.
- Collaborated closely with QA and cross-functional teams to review code, address technical issues and ensure compliance with performance and coding standards.

Education:

- B. Tech in Computer Science and Engineering, India Apr 2005 – Apr 2009

Certifications:

- Terraform Associate HashiCorp Certified.
- AWS Certified Solutions Architect Associate.
- Microsoft Certified Azure Solutions Architect Expert.